NIHR | National Institute for Health and Care Research

Reference number	NIHR209251
Programme	Research for Patient Benefit
Call	RfPB Competition 55 Stage 1
Lead Applicant	Professor Patrick White
Host organisation	Guy's and St Thomas' NHS Foundation Trust
Research Title	Improving safety and TReatment precision for Acute COPD Exacerbations: Feasibility trial of a primary care assessment pathway using point-of-care testing (TRACE).
Start Date	01/11/2025
Grant Duration (Months)	24
End Date	31/10/2027
Estimated Research Cost	£291,325.51
Estimated NHS Support & Treatment costs	£60,519.00
Research Type	Primary Research

Lead Applicant Details

Lead Applicant

Full Name	Professor Patrick White	**Position**	Professor of Primary Care Respiratory Medicine
ORCID iD	0000-0002-2047-8787	**Institution**	King's College London

Specify your (lead applicant) role in the research
Professor White, with Dr Harries as co-lead, will coordinate the research and be responsible for its delivery. He will specifically manage the finances, lead the analysis, and lead scientific reporting and dissemination. Dr Harries is an early career researcher who, in addition to being jointly responsible for the research coordination and delivery, will manage the research team. He will be responsible for recruitment of participating general practices and pharmacies, and the delivery of the research in these research sites. Dr Harries will be mentored by Professor White. Professor White will act as the PPI Lead for the application.

% FTE commitment	10

Joint Lead Applicant

Full Name	Dr Timothy Harries	**Institution**	King's College London
ORCID iD		**Position**	NIHR Academic Clinical Lecturer

Justification for Joint Lead Applicant and role in research
Professor White and Dr Harries have generated the research question and design together, have complementary skills and experience, and will carry through this research jointly to the definitive trial should this feasibility trial be successful. Professor White has established experience as a trialist and was a co-investigator in the trial that demonstrated the utility of using the point-of-care test for C-Reactive Protein in reducing inappropriate prescription of oral corticosteroids COPD in a primary care setting. Dr Harries is an early career (post-doctoral) researcher & GP with established experience investigating the role of eosinophil count in targeting the prevention and treatment of COPD exacerbations. He has also investigated the safety of the withdrawal of inhaled corticosteroids in patients with COPD in a primary care setting. He has been a co-applicant on a successful RfPB grant, recipient of an NIHR DRF award, and has 17 publications . Prof White will provide mentorship/guidance.

% FTE	10

Research Team Details

Co-Applicant	**Organisation**	**Specify Co-Applicant role in research**	**% FTE**
Dr Richard Russell	King's College London	An experienced respiratory researcher, co-investigator on the STARR2 trial, the first to demonstrate safe efficacy of blood eosinophil count point-of-care testing in COPD primary care.	5
Dr Salma Ayis	King's College London	Statistical lead advising on the calculation of the sample size for the main trial, the selection of variables, and the statistical analysis plan.	5
Mr Leslie Hamilton	PPI representative	Mr Hamilton is a member of the KCL COPD PPI group and is an experienced PPI co-applicant, participating in several primary care COPD trials.	5

COI declaration
Please declare any conflicts or potential conflicts of interest that you or your co-applicants may have in undertaking this research, including any relevant personal, non-personal and commercial interest that could be perceived as a conflict of interest.

None of the co-applicants have declared any conflicts or potential conflicts of interest in undertaking this research.	
Agreement to terms and conditions.	☒Confirmed

Plain English Summary of Research

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Chronic obstructive pulmonary disease (COPD) is irreversible and affects breathing and quality of life, reducing life expectancy. Acute “flare-ups”, called acute exacerbations of COPD (AECOPD), worsen symptoms, sometimes causing hospitalization. National Institute for Health and Care Excellence recommends that COPD patients have a home rescue pack of antibiotics and oral corticosteroids (OCS) for AECOPDs. It is unclear when these medications are helpful. Antibiotic use contributes to antibiotic resistance, a major national problem. OCS can cause diabetes, bone thinning, and cataracts.

New blood tests indicate within minutes if these drugs will be helpful, improving AECOPD treatment safety. They are point-of-care tests (POCT). One POCT measures levels of a chemical called C-reactive protein. A high level indicates that antibiotics are needed. A second POCT measures blood eosinophil count. A high level indicates that OCS are needed. Sometimes both drugs are necessary. POCTs stop patients being given these drugs if they are unlikely to help. Clinical trials report that, during AECOPD, three out of four patients have normal C-reactive protein and one out of three patients have normal blood eosinophil counts. POCT use has reduced the prescription of both drugs for AECOPD without harm.

This study will test the feasibility of doing a full clinical trial of the effectiveness of using POCTs during AECOPDs. An AECOPD assessment pathway will be developed and tested with community pharmacists in their pharmacies. General practices and the pharmacies they work with most closely will be recruited. Practices will be allocated at random to an intervention group or a usual care group. Registered COPD patients from each group will be invited to join the study.

A rescue pack drug prescription will be provided for all participants in the pharmacies they attend. COPD patients registered at ‘usual care’ practices will be asked to attend the pharmacy for their rescue pack drugs if they experience a 'flare-up'. COPD patients registered at the ‘intervention practices’ will be asked to attend the pharmacy for a POCT assessment if they experience a 'flare-up'. For patients in intervention practices, test results will determine if one or two of the rescue pack drugs are recommended. Pharmacists will be trained in POCT use and interpretation. Detailed instructions will be developed for pharmacists to follow, including when to seek advice from the participant’s GP.

The feasibility trial will assess: a) if enough practices and pharmacies can be recruited for the full trial; b) the effectiveness and acceptability of pharmacist training; c) the numbers of COPD patients who agree to participate; d) the frequency of AECOPDs in these patients; e) the proportion of patients with flare-ups who find pharmacy attendance acceptable.

This project was developed with advice from a Patient and Public Involvement Group, pharmacists and GPs.

Research Plan

Research Plan

Problem being addressed

Acute exacerbations of COPD (AECOPD) are a major cause of morbidity and mortality in the UK.[1] The National Institute for Health and Care Excellence (NICE) recommends rescue packs of antibiotics and oral corticosteroids (OCS) for all patients following an AECOPD.[1] In some patients their use may prevent progression of a future exacerbation and reduce the risk of hospitalization. Widespread usage of AECOPD rescue packs increases the likelihood of overuse of antibiotics and OCS, with associated risks of harm. AECOPD treatment accounts for 11.5% of antibiotics prescribed in primary care.[2] Adverse effects of antibiotics include antimicrobial resistance; adverse effects of OCS include diabetes, osteoporosis and cataracts.

The decision to prescribe antibiotics and OCS for AECOPD in primary care relies on symptoms and examination alone. Rescue packs are commonly issued without clinical assessment and often obtained directly from the pharmacist to avoid delay. Use in primary care of point-of-care tests (POCTs) of serum C-reactive protein (CRP) and blood eosinophil count (BEC), can target prescription of antibiotics and OCS during AECOPDs and guide their safe and appropriate use.[3,4] POCT usage has reduced prescription of antibiotics by 22% and of OCS by 33% for AECOPD with no evidence of harm. Results from CRP POCT trials suggest they could reduce antibiotic prescribing in AECOPD by up to 76%.[3]

POCTs are valuable for improving antimicrobial stewardship and mitigating OCS-induced adverse effects. They have been adopted rarely in the UK.[5] Infrequent in-person clinical evaluation of AECOPDs and the provision of rescue packs as the default option makes routine POCT use in general practice logistically challenging. Siting a service for POCT delivery in a community pharmacy would provide rapid access in an AECOPD and achieve economies of scale by making the test accessible to patients from different local practices.

Improving public/patient health, well-being and health/care services

This proposal aligns with the DHSC's policy "Confronting antimicrobial resistance 2024-2029" by improving clinical decision support and optimizing antimicrobial use.[6] It is concerned with patient benefit. It proposes a new approach to safety, efficiency and effectiveness of treatment of a vulnerable group of older adults with COPD while improving rapid access when needed.

From the feasibility trial we will learn the willingness and capacity of patients, community pharmacists, and general practices to adopt a more personalized treatment pathway for AECOPDs. We will identify facilitators and obstacles to pathway delivery and the optimization of the context for its implementation. We will provide information for the powering and conduct of a national hybrid cluster randomized controlled trial (RCT) and implementation study of the impact of POCTs delivered in pharmacies. This research will test a model to improve antimicrobial and steroid stewardship in an accessible and cost-effective way.

Evidence review

C-reactive protein (CRP) POCT to guide antibiotic prescription during respiratory tract infection

Anti-microbial resistance is a global challenge with 1.5 million deaths from lower respiratory tract infections (LRTIs) attributed to resistant organisms in 2019.[6] Although antibiotics may reduce treatment failure among hospitalized COPD patients, evidence of their effectiveness in mild or moderate exacerbations is equivocal: a Cochrane review of antibiotic vs. placebo therapy in such patients described the beneficial effects of antibiotics as small or absent.[7] In a minority of AECOPDs, raised blood CRP levels are associated with bacterial pathogens in patients' sputum,[8] but in more than 70% of cases POCT CRP levels in a primary care trial were normal.[3]

During mild/moderate COPD exacerbations, independent predictors of treatment failure without antibiotics are sputum purulence (OR: 6.1) and CRP level >40mg/L (OR: 13.4).[9] The CRP POCT has been recognized as a potentially useful tool to limit LRTI antibiotic prescribing among adults in the community.[10] A primary care pilot study of CRP POCT assessment of RTIs reported a 22% decrease in antibiotic prescriptions for pneumonia from using the test.[11] Testing was acceptable to patients[11] as were pharmacist consultations for RTI management.[12]

A recent Cochrane systematic review of the impact of CRP POCTs on antibiotic prescribing for RTIs in primary care found their use significantly reduced antibiotic prescription at index consultation by 23% (RR 0.77, 95% CI 0.69 to 0.89), without harmful effects.[13] Antibiotic prescription within 28 days of follow-up also fell without harm.

Blood eosinophil count (BEC) POCT to guide OCS prescription during AECOPD

Up to 30% of AECOPD events are associated with Type-2 eosinophilic airways inflammation likely to respond to OCS treatment.[14] Clinical characteristics of an AECOPD do not allow differentiation by type of inflammatory response, illustrating the need for an objective marker to guide OCS use. The blood eosinophil count (BEC) has successfully identified appropriate treatment in RCTs of AECOPDs in hospital, using venous blood samples, and in primary care using the POCT.[4,14] A BEC ≥300 cells/μL helps identify patients likely to benefit from OCS.[15] Although undifferentiated OCS treatment of AECOPD is given widely hoping to accelerate symptom recovery, associations exist in COPD patients between a single OCS course and higher rates of sepsis, venous thromboembolism, fractures and hyperglycaemia.[16] These adverse outcomes emphasize the importance of knowing when OCS rescue courses are unlikely to be beneficial.

Implementation of POCT strategies for assessment of antibiotic prescribing in primary care

NHS England recommends establishing and expanding POCT usage in community pharmacies to improve antibiotic prescription.[17] Community pharmacists' role in undertaking POCTs has been endorsed but their contribution to AECOPD assessment has not been evaluated. This project aligns with the community pharmacy contractual framework reinforcing pharmacists' role in extended healthcare delivery.[18] Feasibility studies of the delivery of CRP POCTs by pharmacists report positive stakeholder and patient opinions.[12] The impact of POCTs in community pharmacies reported in a systematic review showed promising results in reducing risk of inappropriate medication dispensing.[19]

Our project seeks to achieve change in the NHS by implementing NHSE's recommendation to expand the use of POCTs to improve appropriate antibiotic prescription.[18] The current AECOPD assessment in primary care often consists of either a telephone consultation or no clinical review. In these circumstances pharmacists are ideally placed to contribute to AECOPD management providing accessible POCTs for antibiotic and OCS prescribing decisions. A pathway that includes immediately accessible POCTs will ensure that rescue packs will be provided where they are indicated. The GP system is less well positioned to provide immediate access to these key tests.

A search of trial registries including the ISRCTN, Clinicaltrials.gov, and the European Union Clinical Trials Register confirms no past or ongoing studies matching or similar to the proposed research.

Research aims/objectives

Aim:

- To assess the feasibility of conducting a cluster randomized controlled trial (RCT) to evaluate and implement a new assessment pathway for AECOPD that includes rapid biomarker POCTs (C-reactive protein & blood eosinophil count) in

community pharmacies.

Objectives:

- To assess the willingness of community pharmacists and associated general practices to test the new pathway (number of pharmacists and number of general practices required to be approached to recruit ten of each).
- To assess the willingness of COPD patients in the study usual care arm to have 'rescue pack' medication prescriptions held by community pharmacists (rather than being held by themselves) and obtained from the pharmacists at onset of an AECOPD (number of COPD patients required to be approached in each such practice to recruit twenty patients).
- To assess the willingness of COPD patients in the study intervention arm general practices to have 'rescue pack' medication prescriptions held by community pharmacists (rather than being held by themselves) and provided by the pharmacist at the onset of an AECOPD subject to the outcome of a POCT test (number of COPD patients required to be approached in each such practice to recruit twenty patients).
- To assess the number of AECOPDs reported and observed over six months in patients in both arms of the feasibility trial to determine the number of patients that will be required to assess outcomes in a clinical cluster RCT of the new pathway.
- To assess the sample size of clusters (pharmacies & general practices) and the number of participating COPD patients required in each cluster to determine an absolute annual reduction of 22% in antibiotics prescribed for AECOPDs or 33% in OCS prescribed for AECOPDS (as co-primary outcomes) in the intervention arm compared to the usual care (standard-of-care) arm.
- To assess patient-participants' and pharmacists' experiences of the trial

Project plan

Preparatory work

A qualitative interview study, currently in process, is being conducted by co-lead applicant, Dr Tim Harries, who has received a Dunhill Medical Trust Academy Ignition Fund award. The aim of this award is to investigate the facilitators and barriers among pharmacists and primary care staff of implementing a new AECOPD assessment pathway. Results will be available by early February 2025 for incorporation in a stage 2 application to RfPB. The proposed new AECOPD treatment pathway will require changes in the management of AECOPDs and will be influenced by the changing context revealed in this investigation. Its design will use the adaptation framework of Moore et al[20], including their 'five steps' and 'reporting recommendations'. An adaptation team of stakeholders with knowledge of the theory and evidence behind the new pathway, and with contextual knowledge of current practice has been assembled. The team includes pharmacists, GPs, respiratory specialists and a health psychologist. This qualitative study will evaluate the behavioural changes necessary to implement a new AECOPD assessment pathway.

The feasibility trial

This cluster randomized feasibility trial will take place in 10 general practices (each a cluster). Five practices will be randomized to an intervention arm in which COPD patients with an AECOPD will be asked to attend a pharmacy for two POCTs and provision of a rescue pack if indicated. Five practices will be randomized to a control arm, in which COPD patients with an AECOPD will be asked to attend a pharmacy for a rescue pack as part of usual care. 10 clusters, average 150 COPD patients per cluster/practice, will ensure representativeness in the study area of the COPD population of these general practices. They will provide estimates of the frequency of AECOPD in these COPD patients and the rate of receipt of rescue packs. The findings will be extrapolated to the national setting for a definitive trial using established local prevalence statistics of COPD, local healthcare delivery characteristics and context, and a stakeholder group that includes local specialists and patient and public representatives.

Inclusion criteria: Ambulant adults aged 40 years and over with a recorded COPD history who have received a prior treatment for ≥1 exacerbation.

Work package 1: Ethics/Governance/Recruitment/Training

Preparation of ethics and governance materials will start at least three months before formal start of the feasibility trial to minimize delays after approval of funding. Pharmacies and general practices will be recruited in tandem. An algorithm will be developed with pharmacist stakeholders and collaborators, Figure 1, for the initial consultation and the provision of two POCT tests (CRP and BEC) in the intervention arm. It will be supported by a formal delivery manual, graphics & references. The process for pharmacies in the usual care arm will include holding of rescue pack prescriptions and provision to patients with POCTs in response to presentation with suspected AECOPD.

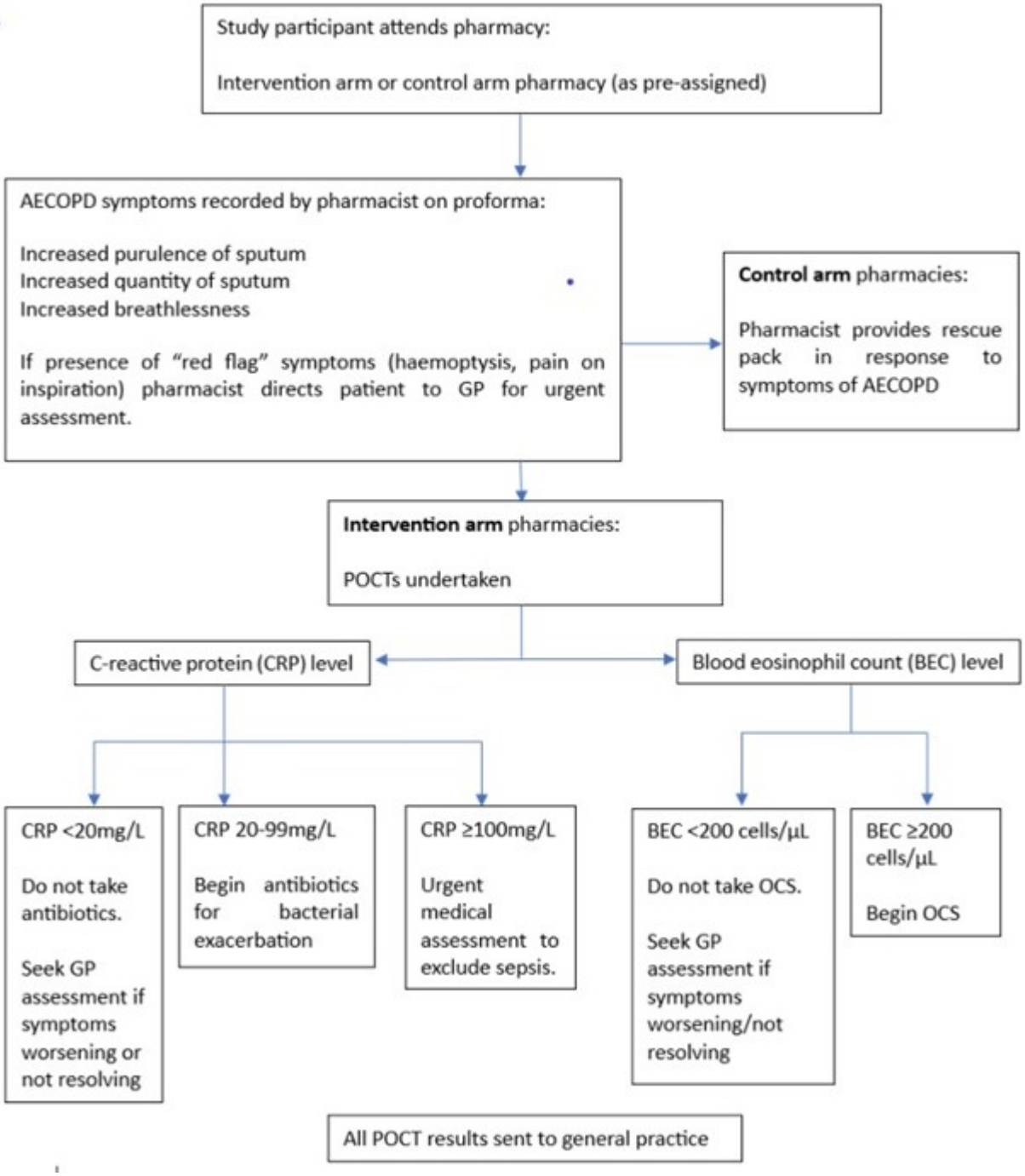

Figure 1: Algorithm for delivery of POCT tests and rescue packs in AECOPD.

Supportive documents, including printed algorithms and intervention manuals will be designed for each practice. These will include processes for receptionists, practice nurses, and GPs, with links to the associated pharmacy. A strategy for recording and prompting of likely AECOPDs in the Electronic Patient Record (EPR) will be developed. Patient materials, including colour leaflets, will be designed for recruitment/support.

Work package 2:

General practices with at least 9,000 patients will be invited to take part. Participating practices must be in proximity to a pharmacist willing to join the feasibility trial with capacity to accommodate and operate two POCT devices. Practices will be randomized to the intervention or control arm. Eligible registered COPD patients will be invited to participate. Upon entering the study, an individual prescription for each drug (antibiotic and OCS) for each enrolled participant will be sent to the designated pharmacy. Eligible COPD patients will have a flag alert placed on their electronic patient record. Attendance at the pharmacy

following the anticipated start of an AECOPD will lead to provision of a rescue pack in usual care patients. Patients in the intervention arm will be given two POCTs (CRP + BEC) and based on the result will be offered the appropriate prescribed medication if indicated.

In the intervention arm, pharmacists will use the algorithm (Figure 1) to classify patients by CRP level (defined as <20mg/L or ≥20mg/L) and BEC level (<200cells/µL or ≥200cells/µL). A copy of the result will be given to them and sent to their registered practice. If the POCT test results are in the normal range (CRP <20mg/L & BEC <200µg/L) for both biomarkers,[3,4] then the participant will be advised not to use either antibiotic or OCS but to seek help should symptoms worsen. If both, or either, biomarker levels are raised, the participant will be advised to take the appropriate rescue pack component(s). If the participant is not improving, they will be advised to seek assessment with their primary care clinician. Those with a CRP ≥100mg/L will be sent for urgent medical assessment to exclude sepsis.

Outcome measures

Data on participants will be collected by general practices.

The **primary** outcome will be the proportion of registered COPD patients who consent to trial participation.

The **secondary** outcomes relating to the objectives above are:

- Proportion of practices approached agreeing to randomization
- Proportion of pharmacists approached that agree to participation
- Rate of COPD exacerbations among patient participants
- Proportion of eligible participants with an AECOPD who attend the pharmacy
- Proportion of AECOPD episodes in the intervention arm assessed by POCT
- Proportion of participants prescribed an antibiotic or OCS after presentation with AECOPD at practice or pharmacy
- Proportion of participants needing a further consultation (primary or secondary care) within two weeks
- Patient-participants' and pharmacists' experience of the trial will be surveyed by questionnaire

Analysis will include frequencies and proportions. Rates of attendance, POCT tests carried out, and drugs prescribed will be compared between intervention and usual care arms using appropriate tests.

Sample sizes

1. **For feasibility trial.** The estimated rate of AECOPD (England) is 0.5-0.625/COPD patient/year. COPD prevalence in England between 1.1% (London) and 5% (North-East) is approximately 4-5%.[21] 2.24-2.8 million COPD patients in England account for 1.4m GP consultations/year.[22] In 10 enrolled London general practices (minimum 9,000 patients/practice), about 1500 COPD patients will experience between 375-470 exacerbations/6-months. These exacerbations will have the potential to be assessed in pharmacies. We estimate a recruitment rate of at least 20% of patients (300 patients experiencing 75-94 AECOPD) from 10 clusters. During 80% of AECOPD episodes, participants are expected to attend the pharmacy for assessment (60-75 AECOPD assessments). With 10 clusters (minimum 20 participants/cluster), at an intra-cluster correlation coefficient of 0.056, an overall 20% recruitment rate may be estimated with a margin of error of ±10%.[23]

2. **For main trial.** With a sample size of 160 in both groups, obtained by sampling at least 8 clusters with 20 subjects each, differences in the two co-binary primary endpoints, antibiotic use and oral steroid use, with reductions of 25% to 30% in each in the intervention group compared to no change or a minimal change of 5% in the control group, will be detected with a power of 90%. This assumes an intra-cluster correlation (ICC) of 0.056, and a significance level of 0.025, using an equally weighted method for testing the two endpoints. These are conservative estimates based on available evidence described above to be confirmed in the feasibility study, and not allowing for dropout.

Data analysis will include generation of sample size data for the future RCT based on recruitment of pharmacies, practices, patients.

Dissemination

To include scientific papers, conference presentations, papers in professional organization publications, and a conference for stakeholders including patient representatives, national charities, pharmacy and general practice organizations, NHS/DOH, and

policy and academic leaders in specialist primary care, pharmacy, respiratory care and health psychology .

The team

The co-applicants include primary care and respiratory academics who have published trials evaluating the impact of POCTs on prescribing of antibiotics and OCS in AECOPD, withdrawal of inhaled corticosteroids in COPD, and the role of eosinophils in guiding treatment in COPD. One of the co-applicants is a PPI representative with experience of previous involvement as a Co-I in a clinical trial of COPD. An established COPD-specific PPI group will support the study.

Timetable

A Gannt chart detailing the study timetable has been uploaded.

Equity, diversity and inclusion (EDI)

COPD prevalence is associated with socioeconomic deprivation[24] and lower educational attainment[25]. COPD patients are more likely to live in rented accommodation and may be slow to seek help or to acknowledge their unmet needs.[26] This research intends to improve the quality and speed of access to diagnostic tests during an AECOPD among this deprived population. The new assessment pathway for COPD patients will be of benefit in reducing the risk of harm from inappropriate medication for patients more likely to be older and more deprived than the general population.

PPI

Our PPI co-applicant Mr Leslie Hamilton has played an important role in the design of the project and in reviewing the research plan and he has provided feedback on the likely acceptability of the intervention to participants. We have met with the King's College London COPD Patient and Public Involvement (PPI) Primary Care Group. The group support using a POCT to target prescribing for AECOPDs to reduce avoidable harms. They support the participation of patients. One patient noted a potential conflict between the result of the POCT and her own perceived need for components of a rescue pack. Her concern highlighted the importance of this research in providing personalized decision-making that would strengthen the safety of AECOPD prescribing.

Non-co-applicant stakeholders

Prof Mona Bafadhel, Principal Investigator in the first trial of the effectiveness of the BEC POCT in reducing OCS prescriptions in primary care (highlighted by "NIHR Evidence").[4]

Mr Hiten Patel, Community Pharmacy policy.

Mr Dilip Joshi, Community Pharmacy academic.

Dr Alison Wright, Health Psychologist, specialist in behavioural change technique research.

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Uploads

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GANNT chart

Improving safety and TReatment precision for Acute COPD Exacerbations: Feasibility trial of a primary care assessment pathway using point of care testing (TRACE).

Start

	2025			2026												2027									
Month of project	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Calendar months	Aug -Oct	N	D	J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O
Feasibility trial																									
Research Ethics and Governance																									
Trial steering group meetings																									
Patient advisory group																									
Prepare research materials																									
Recruit practices & pharmacies for study																									
Identification & consent of eligible participants																									
Participant follow-up (6-months/ participant)																									
Data analysis & write-up																									